

星际介质天文学作业

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23.1 (a) 在 $a_{\max} < 2\lambda/\pi$ 时, 单个氢原子在波长 λ 处的消光截面 $\sigma_{\text{ext}}(\lambda)$ 可以推导如下:

$$\begin{aligned}\sigma_{\text{ext}}(\lambda) &= \int_0^{a_{\max}} \pi a^2 \cdot Q_{\text{ext}}(a, \lambda) \cdot \left(\frac{1}{n_H} \cdot \frac{dn}{da} \right) da \\ &= \int_0^{a_{\max}} 2\pi a^2 \left(\frac{\pi a}{2\lambda} \right)^\beta \cdot \frac{A_0}{a_0} \left(\frac{a}{a_0} \right)^{-p} da \\ &= 2\pi A_0 a_0^{p-1} \left(\frac{\pi}{2\lambda} \right)^\beta \int_0^{a_{\max}} a^{2+\beta-p} da \\ &= 2\pi A_0 a_0^{p-1} \left(\frac{\pi}{2\lambda} \right)^\beta \frac{a_{\max}^{3+\beta-p}}{3+\beta-p} \\ &= \frac{2\pi A_0 a_0^2}{3+\beta-p} \left(\frac{\pi a_0}{2\lambda} \right)^\beta \left(\frac{a_{\max}}{a_0} \right)^{3+\beta-p}\end{aligned}$$

即:

$$\frac{\sigma_{\text{ext}}(\lambda)}{\pi A_0 a_0^2} = \frac{2}{3+\beta-p} \left(\frac{\pi a_0}{2\lambda} \right)^\beta \left(\frac{a_{\max}}{a_0} \right)^{3+\beta-p}$$

代入数据得:

$$\frac{\sigma_{\text{ext}}(\lambda_V)}{\pi A_0 a_0^2} = \frac{2}{3+1.5-3.5} \left(\frac{\pi \times 0.1 \mu\text{m}}{2 \times 0.55 \mu\text{m}} \right)^{1.5} \left(\frac{0.25 \mu\text{m}}{0.1 \mu\text{m}} \right)^{3+1.5-3.5} = 0.763$$

23.1 (b) 由于:

$$\sigma_{\text{ext}}(\lambda) = \frac{2\pi A_0 a_0^2}{3+\beta-p} \left(\frac{\pi a_0}{2\lambda} \right)^\beta \left(\frac{a_{\max}}{a_0} \right)^{3+\beta-p} \propto \lambda^{-\beta}$$

则:

$$\frac{\sigma_{\text{ext}}(\lambda_B)}{\sigma_{\text{ext}}(\lambda_V)} = \left(\frac{\lambda_B}{\lambda_V} \right)^{-\beta}$$

代入数据得:

$$\frac{\sigma_{\text{ext}}(\lambda_B)}{\sigma_{\text{ext}}(\lambda_V)} = \left(\frac{0.44 \mu\text{m}}{0.55 \mu\text{m}} \right)^{-1.5} = 1.398$$

23.1 (c) 由 $A(\lambda) = 1.086\tau(\lambda) = 1.086n\sigma_{\text{ext}}(\lambda)l$ 得:

$$\begin{aligned}R_V &= \frac{A_V}{A_B - A_V} \\ &= \frac{\tau_V}{\tau_B - \tau_V}\end{aligned}$$

$$\begin{aligned}
&= \frac{\sigma_{\text{ext}}(\lambda_V)}{\sigma_{\text{ext}}(\lambda_B) - \sigma_{\text{ext}}(\lambda_V)} \\
&= \frac{1}{\sigma_{\text{ext}}(\lambda_B)/\sigma_{\text{ext}}(\lambda_V) - 1} \\
&= \frac{1}{1.398 - 1} \\
&= 2.515
\end{aligned}$$

23.1 (d) 在 $a_{\text{max}} > 2\lambda/\pi$ 时, 单个氢原子在波长 λ 处的消光截面 $\sigma_{\text{ext}}(\lambda)$ 需要考虑 $[0, 2\lambda/\pi]$ 和 $[2\lambda/\pi, a_{\text{max}}]$ 这两部分的贡献, 分别有不同的形式:

$$\begin{aligned}
\sigma_{\text{ext}}(\lambda) &= \int_0^{2\lambda/\pi} \pi a^2 \cdot Q_{\text{ext}}(a, \lambda) \cdot \left(\frac{1}{n_H} \cdot \frac{dn}{da} \right) da + \int_{2\lambda/\pi}^{a_{\text{max}}} \pi a^2 \cdot Q_{\text{ext}}(a, \lambda) \cdot \left(\frac{1}{n_H} \cdot \frac{dn}{da} \right) da \\
&= \int_0^{2\lambda/\pi} 2\pi a^2 \left(\frac{\pi a}{2\lambda} \right)^\beta \cdot \frac{A_0}{a_0} \left(\frac{a}{a_0} \right)^{-p} da + \int_{2\lambda/\pi}^{a_{\text{max}}} 2\pi a^2 \cdot \frac{A_0}{a_0} \left(\frac{a}{a_0} \right)^{-p} da \\
&= 2\pi A_0 a_0^{p-1} \left(\frac{\pi}{2\lambda} \right)^\beta \int_0^{2\lambda/\pi} a^{2+\beta-p} da + 2\pi A_0 a_0^{p-1} \int_{2\lambda/\pi}^{a_{\text{max}}} a^{2-p} da \\
&= 2\pi A_0 a_0^{p-1} \left(\frac{\pi}{2\lambda} \right)^\beta \frac{(2\lambda/\pi)^{3+\beta-p}}{3+\beta-p} + 2\pi A_0 a_0^{p-1} \left[\frac{a_{\text{max}}^{3-p}}{3-p} - \frac{(2\lambda/\pi)^{3-p}}{3-p} \right] \\
&= \frac{2\pi A_0 a_0^2}{3+\beta-p} \left(\frac{2\lambda}{\pi a_0} \right)^{3-p} + \frac{2\pi A_0 a_0^2}{3-p} \left[\left(\frac{a_{\text{max}}}{a_0} \right)^{3-p} - \left(\frac{2\lambda}{\pi a_0} \right)^{3-p} \right]
\end{aligned}$$

即:

$$\frac{\sigma_{\text{ext}}(\lambda)}{\pi A_0 a_0^2} = \frac{2}{3+\beta-p} \left(\frac{2\lambda}{\pi a_0} \right)^{3-p} + \frac{2}{3-p} \left[\left(\frac{a_{\text{max}}}{a_0} \right)^{3-p} - \left(\frac{2\lambda}{\pi a_0} \right)^{3-p} \right]$$

23.1 (e) 对于 $\lambda_V = 0.55 \mu\text{m}$, $a_{\text{max}} = 0.35 \mu\text{m} < 2\lambda_V/\pi = 0.35014 \mu\text{m}$, 则:

$$\begin{aligned}
\frac{\sigma_{\text{ext}}(\lambda_V)}{\pi A_0 a_0^2} &= \frac{2}{3+\beta-p} \left(\frac{\pi a_0}{2\lambda_V} \right)^\beta \left(\frac{a_{\text{max}}}{a_0} \right)^{3+\beta-p} \\
&= \frac{2}{3+1.5-3.5} \left(\frac{\pi \times 0.1 \mu\text{m}}{2 \times 0.55 \mu\text{m}} \right)^{1.5} \left(\frac{0.35 \mu\text{m}}{0.1 \mu\text{m}} \right)^{3+1.5-3.5} \\
&= 1.068
\end{aligned}$$

对于 $\lambda_B = 0.44 \mu\text{m}$, $a_{\text{max}} = 0.35 \mu\text{m} > 2\lambda_B/\pi = 0.28011 \mu\text{m}$, 则:

$$\begin{aligned}
\frac{\sigma_{\text{ext}}(\lambda_B)}{\pi A_0 a_0^2} &= \frac{2}{3+\beta-p} \left(\frac{2\lambda_B}{\pi a_0} \right)^{3-p} + \frac{2}{3-p} \left[\left(\frac{a_{\text{max}}}{a_0} \right)^{3-p} - \left(\frac{2\lambda_B}{\pi a_0} \right)^{3-p} \right] \\
&= \frac{2}{3+1.5-3.5} \left(\frac{2 \times 0.44 \mu\text{m}}{\pi \times 0.1 \mu\text{m}} \right)^{3-3.5} + \frac{2}{3-3.5} \left[\left(\frac{0.35 \mu\text{m}}{0.1 \mu\text{m}} \right)^{3-3.5} - \left(\frac{2 \times 0.44 \mu\text{m}}{\pi \times 0.1 \mu\text{m}} \right)^{3-3.5} \right] \\
&= 1.447
\end{aligned}$$

则 R_V 为:

$$\begin{aligned}
R_V &= \frac{A_V}{A_B - A_V} \\
&= \frac{\tau_V}{\tau_B - \tau_V}
\end{aligned}$$

$$\begin{aligned}
&= \frac{\sigma_{\text{ext}}(\lambda_V)}{\sigma_{\text{ext}}(\lambda_B) - \sigma_{\text{ext}}(\lambda_V)} \\
&= \frac{1}{\sigma_{\text{ext}}(\lambda_B)/\sigma_{\text{ext}}(\lambda_V) - 1} \\
&= \frac{1}{1.447 \div 1.068 - 1} \\
&= 2.823
\end{aligned}$$